

Identifying the off-diagonal "exchange scalar": for TECT's local

TECT verification-first repository · claim B2-PROPA-HLAYER

first issued 2026-06-10 · this version issued 2026-06-10 · v1.0

1. Purpose and scope

T-018 reduced the full off-diagonal operator-norm residual to a scalar $b_{\text{exch}}(Q) < b_*(Q)$, the E-normalised "Fock exchange" entry, inheriting the characterisation from hdiag-full-operator-norm-formulation v1.0: "an A-independent exchange block whose sign is not fixed by $u_{\text{eff}} > 0$ ". This note IDENTIFIES that object precisely and shows it is not a new unbounded residual. Scope: the mean-field (Gaussian–Hartree) off-diagonal Hessian at the production anchor $\mu^2 = 5 \times 10^{-3}$; the reconciliation with Math428's explicit Bloch computation. No tier action.

2. TECT's interaction is local; the mean-field functional is a density functional

The TECT action carries a CONTACT quartic ($u = -0.86$) and contact sextic ($v = +3.24$). At the Gaussian–Hartree level the interaction free energy is the local density functional

$$\Phi[G] = \int_x f(M(x)), \quad f(M) = \frac{3u}{4}M^2 + \frac{5v}{2}M^3, \quad M(x) = G(x, x),$$

the Wick factors ($\langle \phi^4 \rangle = 3G^2$, $\langle \phi^6 \rangle = 15G^3$) already counting all contractions. For a contact interaction the exchange (Fock) integral $\iint \delta(x-y)G(x, y)^2 = \int_x G(x, x)^2$ coincides with the direct (Hartree) term: Hartree = Fock, with no separate sign-indefinite exchange at mean-field order.

3. The off-diagonal Hessian is the stabilising density-density term

Write $M(x) = M + \sum_{Q \neq 0} \rho_Q e^{iQx}$, $\rho_Q = \int_q \delta G(q, q+Q)$. The second variation of the local functional in the off-diagonal sector is

$$\frac{\delta^2 \Phi}{\delta \rho_Q \delta \rho_{-Q}} = f''(M) = \frac{3u}{2} + 15vM = \frac{3}{2}(u + 10vM) = \frac{3}{2}u_{\text{eff}} \quad (\text{INDEPENDENT of } Q)$$

which at the anchor is $\frac{3}{2}(2.685) = +4.03 > 0$. Three consequences: (a) it is *Q-independent* (verified numerically to 3.6×10^{-10} over $Q = 1, \dots, 7$): a pure density-density coupling, diagonal in Q ; (b) it is *strictly positive* (STABILISING) – there is NO A-independent attractive Fock exchange; (c) it equals the $l = 0$ breathing-sector curvature of T-016, the same local-functional second derivative $f''(M)$ governing both. The full-operator-norm note's "sign not fixed by $u_{\text{eff}} > 0$ " conflated the BARE $u = -0.86 < 0$ with this DRESSED Hessian, which carries $u_{\text{eff}} = +2.685 > 0$.

4. The genuine off-diagonal obstruction (A-dependent + two-loop)

With the A-independent exchange shown stabilising, the off-diagonal obstruction is two already-tracked objects, NOT a new scalar:

- (i) **A-dependent condensate bubble.** Math428's explicit Bloch computation gives the off-diagonal share $-\frac{1}{4} \sum_{Q \neq 0} W(Q)^2 B(|Q|)$ with $W(Q) = (3u + 30vM_R)A^2 p_2(Q) + 5vA^4 p_4(Q)$, i.e. $O(A^4)$, condensate-induced, diagonal in Q , and R_{lead} -controlled (≤ 0.650 class-wide, T-018). The enumerated Bogoliubov bands are POSITIVE even at the no-resummation-credit value $\rho = 1$ (minimum $+6.7 \times 10^{-4}$; calibrated $\rho = 0.415$ is a further credit).
- (ii) **Beyond-mean-field two-loop exchange.** The genuine non-local $\iint G(x, y)^2$ exchange appears at TWO LOOPS (the fluctuation/sunset correction), i.e. the third cumulant – exactly the SC-SCOPE object, bounded by the sunset cap $\times 1.13$ and lifted@thin-certified (2026-06-09).

Thus the T-018 residual is REFRAMED: not "bound an unbounded A-independent b_{exch} ", but "extend (i) the R_{lead} control and (ii) the SC-SCOPE two-loop bound class-wide to non-enumerated patterns" – the open RES-5.

5. Status and honest residual

T4 STRONG EVIDENCE (reframing). Established: for the local interaction the A-independent off-diagonal Hessian is $+\frac{3}{2}u_{\text{eff}} > 0$ (stabilising, Q-independent); there is no A-independent attractive Fock exchange. The off-diagonal obstruction is the A-dependent R_{lead} -controlled bubble (T-018) plus the beyond-mean-field two-loop = SC-SCOPE (thin). **OPEN (RES-5):** the class-wide extension of the resummation ρ and the two-loop bound to non-enumerated patterns (Math428 verified the five enumerated readings, positive even at $\rho = 1$). Anchored: $u_{\text{eff}} > 0$ is the ROBUSTNESS-MU2 fact ($[\times 0.5, \times 2]$ closed). No tier flip: B2 T6, B1 T6 on {H-LAYER}.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) *"This contradicts the full-operator-norm note's exchange residual."* PARTLY: it CORRECTS the identification. The note's "A-independent Fock exchange, sign unfixed" is, for a LOCAL functional, the stabilising density-density $+\frac{3}{2}u_{\text{eff}}$; the genuine non-local $\iint G^2$ exchange is a two-loop (SC-SCOPE) object, not mean-field. The reframing is the value: the residual is already-tracked ($R_{\text{lead}} + \text{SC-SCOPE}$), not a new scalar.
- (β) *"Hartree=Fock only for a strictly contact interaction."* VALID and used: the TECT quartic/sextic are contact; the kinetic $K_0(q) = c(q^2 - q_0^2)^2$ carries the momentum structure but the INTERACTION vertices are local, so the mean-field exchange reduces to the local $\int f(M)$. A non-local interaction would reinstate a separate Fock term; TECT's does not.
- (γ) *"Does this close the off-diagonal norm or flip B1?"* DISMISSED: no. The class-wide ρ and the two-loop bound for non-enumerated patterns remain open (RES-5). This note removes a MIS-identified residual (the A-independent exchange) and routes the real one onto SC-SCOPE; it does not close it. B1/B2 stay T6.

Quantitative sanity check (mandatory): *Q-independence* – the off-diagonal Hessian deviates from $f''(M)$ by 3.6×10^{-10} over $Q = 1 \dots 7$ (machine zero, the local functional gives a pure $|\rho_Q|^2$ form);

$sign - f''(M) = \frac{3}{2}u_{\text{eff}} = +4.03 > 0$ (bare $u = -0.86 < 0$ is irrelevant; the dressed $u_{\text{eff}} > 0$ enters);
consistency – equals the T-016 $l = 0$ curvature; *Math428* – off-diagonal $O(A^4)$, bands $+6.7 \times 10^{-4} > 0$
at $\rho = 1$.

7. Result footer

Result ID:	B2-PROPA-HLAYER / off-diagonal exchange-scalar identification
Precise statement:	TECT's interaction is LOCAL, so the mean-field free energy is a density functional $\Phi = \int f(M(x))$, $f(M) = (3u/4)M^2 + (5v/2)M^3$. Its off-diagonal ($Q! = 0$) Hessian is $f''(M) = (3/2)u_{\text{eff}} > 0$, Q-INDEPENDENT (verified $3.6e-10$): the DIRECT density-density term, STABILISING, NOT an attractive Fock exchange (for a local functional Hartree = Fock; the note's "sign-unfixed exchange" conflated bare $u < 0$ with the dressed $u_{\text{eff}} > 0$). The genuine off-diagonal obstruction is (i) the A-dependent condensate bubble (Math428, $O(A^4)$, $R_{\text{lead}} \leq 0.650$ class-wide; bands $+6.7e-4 > 0$ at $\rho = 1$) and (ii) the beyond-mean-field two-loop int-int G^2 exchange = sunset = SC-SCOPE (cap $\times 1.13$, lifted@thin-certified). The T-018 b_{exch} is NOT a new unbounded scalar.
Scope:	Mean-field (Gaussian-Hartree) off-diagonal Hessian; anchor $\mu^2 = 0.005$ ($u_{\text{eff}} > 0 = \text{ROBUSTNESS-MU2}$). Class-wide $\rho +$ two-loop for non-enumerated patterns = open RES-5.
Dependencies:	hdiag-full-operator-norm-formulation (the residual this reframes); hdiag-offdiag-gershgorin-threshold v1.1 (T-018, $R_{\text{lead}} \leq 0.650$); Math428 (explicit Bloch, $O(A^4)$, ρ , bands); isotropy-infimum-core v1.1 (T-016, $f''(M)$ consistency); SC-SCOPE (two-loop sunset).
Evidence grade:	ANALYTIC (local-functional Hessian) + EXECUTED (res5_019_exchange_scalar_identification.py 5/5) + INHERITED (Math428 bands; SC-SCOPE thin closure).
Reproduction command:	python codes/vacuum/res5_019_exchange_scalar_identification.py
Expected output:	$f''(M) = (3/2)u_{\text{eff}} = 4.03 > 0$, Q-independent (dev $3.6e-10$); Math428 band $\min(\rho = 1) = +6.7e-4$; 5/5 PASS.
Falsification gate:	A non-local exchange term in the MEAN-FIELD functional (would reinstate a separate Fock block); or $f''(M) \leq 0$ at the anchor ($u_{\text{eff}} \leq 0$, refuted: $u_{\text{eff}} = +2.685$); or a Math428 band < 0 at $\rho = 1$.
Tier before / after:	No tier flip. B2 T6, B1 T6 on {H-LAYER}. The off-diagonal residual is reframed: the A-independent exchange is stabilising ($+3/2 u_{\text{eff}}$), the genuine residual is R_{lead} (T-018) + the two-loop = SC-SCOPE.
No-overclaim statement:	Does NOT close the off-diagonal operator norm and does NOT flip B1/B2. It corrects the IDENTIFICATION of the T-018 residual: the A-independent exchange is stabilising (no residual), the genuine residual is the A-dependent R_{lead} -controlled bubble plus the beyond-mean-field two-loop = SC-SCOPE (thin). Class-wide extension to non-enumerated patterns is open RES-5. B1/B2 stay T6.
Next required action:	Extend (i) the $R_{\text{lead}} < 1$ bound and (ii) the SC-SCOPE two-loop (sunset) thin closure class-wide to non-enumerated admissible patterns = the open RES-5; the mean-field A-independent sector is settled (stabilising).